

CUET IEEE COMPUTER SOCIETY · STUDENT BRANCH CHAPTER

presents

ZERO TO PRODUCTION

PHASE 2 · THE ULTIMATE HACKATHON

The IEEECS Engineering Challenge

LEARN IT > BUILD IT > SHIP IT

OFFICIAL RULEBOOK

Saturday, 8 August 2026 · 9:00 AM – 8:00 PM · CUET Campus

POWERED BY

Poridhi.io

Technical Partner · Official Judging Panel

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1 THE EVENT

Zero to Production is the engineering-focused hackathon at the heart of the IEEECS Engineering Challenge. It is a single-day, on-campus build sprint in which teams take a problem statement they have never seen before and turn it into a working, containerized, deployed, documented system before the clock runs out.

Most hackathons reward the best demo. This one rewards the best engineering. Over eleven hours your team will move through the same loop a real product team moves through: read the requirements, design the system, write the code, test it, containerize it, wire up a pipeline, push it to a live environment, and then stand in front of engineers and defend every decision you made.

Teams are free to choose any technology stack, language, database, or cloud platform, so long as the final solution satisfies the announced problem statement. There is no prescribed framework and no template to fill in. How you get to production is your call; that you get there is what we are measuring.

PARTNERSHIP

Zero to Production is organised in partnership with Poridhi.io, who serve as the event's Technical Partner and Official Judging Panel.

Every submitted solution will be reviewed and evaluated by working engineers from the Poridhi.io team, not by a generic panel. The feedback you receive is industry feedback.

2 WHAT MAKES THIS HACKATHON DIFFERENT

If you have joined a hackathon before, set your expectations aside. Four things make this event unlike the rest.

A finished feature list is not the goal

A polished front end sitting on top of a fragile, untested, undeployable codebase will not win here. A slightly smaller feature set that is modular, tested, containerized and running on a public URL will. Judges open your repository, read your commits, run your Docker build, and hit your deployed endpoint. Build accordingly.

You will ship to a real environment

Deployment is not a bonus round. Getting your stack live is part of the core requirement, and a submission that never leaves localhost has left points on the table. This is the single skill that most separates students from working engineers, and it is the skill this event is built around.

You will have to defend your work

Every team presents to the judging panel and answers questions about their architecture. Why this database? Why did you split those services? What breaks first under load? What would you fix with another day? Teams that understand their own system score well. Teams that cannot explain it do not, no matter how much code is in the repo.

AI is a tool you are expected to use well

AI-assisted development is fully permitted and actively encouraged, including Poridhi's Puku Editor and Puku CLI. Agentic coding is part of how modern engineering gets done, and using it well is a skill in its own right. The line we draw is understanding: whatever generated your code, you own it, and you must be able to explain it.

WHAT YOU WALK AWAY WITH

A deployed, production-shaped project and a public repository with real CI history — something you can put in front of a recruiter.

Direct code-level feedback from the Poridhi.io engineering team.

Access to the complete Phase 1 learning content and Poridhi's hands-on lab facilities.

A Certificate of Participation for every registered participant, regardless of placement.

3 EVENT SUMMARY

ITEM	DETAILS
Event	Zero to Production — Phase 2: The Ultimate Hackathon
Date	Saturday, 8 August 2026
Time	9:00 AM – 8:00 PM (check-in from 8:30 AM)
Format	On-campus, in person, single day
Venue	CUET Campus — [VENUE / ROOM TO BE CONFIRMED]
Team Size	1 – 3 members
Eligibility	Open for all students; cross-batch and cross-department teams are permitted
Registration Fee	1 member: 300 BDT 2 members: 500 BDT 3 members: 600 BDT
Registration Deadline	6 August 2026, 11:59 PM (extended)
Technical Partner & Judge	Poridhi.io
Problem Statement	Revealed live at the Opening Ceremony

Phase 1 attendance is not a prerequisite. Every registered team receives the Phase 1 learning content and lab access on confirmation, so a team joining at Phase 2 can still prepare before event day.

4 REGISTRATION

The registration deadline has been extended to 11:59 PM on 6 August 2026. Late registrations and late payments will not be accepted, and a form submitted without a matching payment is not a confirmed registration.

STEP	DETAILS
1. Register online	Submit the official registration form at https://forms.gle/5VEtWzPyauEWQa4k9 with your team name and all member details.
2. Pay the fee	Send the applicable fee via bKash to 01837603873. Use your team name as the payment reference.
3. Confirm	Keep your transaction ID. Your team is confirmed once the payment is matched to your form submission.
Deadline	All forms and payments must be completed by 6 August 2026, 11:59 PM.

Every registered participant receives:

- Access to the complete Phase 1 learning content
- Access to Poridhi's hands-on lab facilities, including Docker and AWS lab tracks
- Access to the Puku Editor and Puku CLI for the duration of the event
- A Certificate of Participation

NOTE ON FEES

The registration fee covers venue logistics, meals for the day (light breakfast, lunch and evening snacks) and event operations.

Fees are generally non-refundable once a registration has been confirmed. If you have a genuine reason you can no longer attend, contact the organisers before the event and we will review it case by case.

5 EVENT DAY TIMELINE

All times are tentative and will be confirmed at the Opening Ceremony. Announcements during the day are made through the official event channel; keep it open.

TIME	SESSION	NOTES
08:30 – 09:00	Check-in, setup & light breakfast	Arrive early. Set up your machines, connect to Wi-Fi, verify tooling.
09:00 – 09:30	Opening ceremony & problem reveal	The problem statement is announced. Repositories are created after this point, not before.
09:30	Build window opens	The clock starts.
12:30 – 13:30	Lunch (full meal, rolling)	Take it in shifts; the clock does not stop.
14:00	Mentor checkpoint	Optional. Walk a mentor through your design so far and get course-corrected early.
16:00	[OPTIONAL: MID-EVENT CONSTRAINT DROP]	If used: an additional requirement is released that must be integrated into your existing build.
16:30	Evening snacks	Light refreshments served at your workstation. No break required.
17:30	Deployment reminder	Begin deploying now if you have not. Do not leave shipping to the final hour.
18:30	Code freeze & submission deadline	Final commits and submission form. Commits after this timestamp are not evaluated.
18:30 – 19:45	Team presentations & judging	Each team presents and takes questions from the Poridhi.io panel.
19:45 – 20:00	Closing & wrap-up	Certificates of Participation distributed. Winners are NOT announced today.

MANAGE THE CLOCK, NOT JUST THE CODE

The most common way teams lose points here is spending nine hours on features and one hour on Docker, CI and deployment. Reverse that instinct.

A workable plan: design and scope by 10:30, core functionality by 14:30, containerized by 16:00, deployed by 17:30, documentation and rehearsal in the final hour.

6 RULES & REGULATIONS

6.1 Team Composition

- Each team must consist of one to three members. Solo entries are permitted and compete on the same terms.
- The event is open to all students. Teams formed across different batches and departments are permitted.
- Every participant may belong to only one team.
- Team composition is locked at the registration deadline. Substitutions after that point require organiser approval.

6.2 Project & Code Rules

- Teams must create a new GitHub repository after the Opening Ceremony and use it for all submitted work.
- The repository must be public at the time of submission. Private repositories will not be evaluated.
- The problem statement is revealed only at the Opening Ceremony, so the solution itself is built entirely on the day. Everything around the solution can and should be prepared in advance: install your tooling, configure your cloud accounts, set up your editor, agree on your team's workflow.
- Standard project scaffolding is allowed. Framework generators, official starter templates, UI component libraries, open-source packages and public APIs are all fine.
- Commit early and commit often. Your commit history is the record of how your system took shape, and it is one of the first things judges look at.
- Every external library, API, dataset or service used must be acknowledged in the project README.
- Secrets, API keys and credentials must not be committed. Use environment variables; this is part of the code quality score.

6.3 AI Tool Usage Policy

AI-assisted tools may be used freely for development, debugging, testing, documentation and learning throughout the event. This includes Poridhi's Puku Editor and Puku CLI, and any other assistant your team prefers.

We do not treat AI usage as a shortcut to be policed. We treat it as tooling, and using it skilfully is part of modern engineering practice. What matters is what you can account for.

IMPORTANT

AI tools may assist, but they do not replace understanding. Every team member must be able to explain the team's architectural decisions and implementation choices to the judges.

If a team cannot explain how a substantial part of its own system works, that section is treated as unverified and scored accordingly.

6.4 Fair Play & Disqualification

The following will result in score penalties or disqualification at the organisers' discretion:

- Copying another team's code, or sharing your own solution with another team during the event.
- Submitting a project as original work when it is a fork or clone of an existing repository without substantial, disclosed original work.
- Falsifying commit timestamps or history.
- Interfering with another team's work, environment, network or hardware.
- Misrepresenting a deployment, for example presenting a mocked interface as a live working system.
- Any breach of the Code of Conduct in Section 13.

Organiser and judge decisions on all of the above are final.

7 JUDGING & SCORING

Submissions are evaluated by the Poridhi.io engineering panel against the breakdown below. Read this section before you start building; it is the clearest statement of what this event values.

7.1 Scoring Breakdown

CRITERION	WHAT THE JUDGES LOOK FOR	WEIGHT
System Architecture & Design	Sensible service and module boundaries, clean API design, appropriate data modelling, decisions that would survive the system growing.	25
Functionality & Completeness	How much of the announced problem statement actually works end to end, and how reliably.	25
Code Quality & Testing	Readable, organised code. Input validation, error handling, no hardcoded secrets, unit tests over core logic.	15
Containerization & CI	A working Dockerfile and Compose setup. A functioning CI pipeline that builds and tests on push.	15
Deployment & Production Readiness	The application reachable at a live URL, with the full stack brought up cleanly and repeatably.	10
Documentation	A README that lets a stranger understand, run and deploy the project: architecture notes, setup, deployment, API reference.	5
Presentation & Defence	Clarity of the demo and the quality of answers to the panel's technical questions.	5
TOTAL		100

PARTIAL CREDIT IS REAL

You are not scored pass or fail on each row. A half-finished feature set that is properly containerized, tested and deployed will outscore a long feature list with none of that engineering around it.

If you are running out of time, protect the pipeline and the deployment before you add one more feature.

7.2 Presentation & Defence

Each team gets a short slot with the judging panel at the end of the day. Expect roughly five minutes to demonstrate the working system and explain your architecture, followed by questions.

You do not need slides. What helps is a live demo, your architecture on a whiteboard or a single diagram, and a clear-eyed account of what you built, what you cut, and what you would do next. Every member should be prepared to answer, not only the team lead.

Questions the panel is likely to ask:

- Why did you draw your service or module boundaries there?
- Walk us through what happens on this request, end to end.
- What is the first thing that breaks if traffic multiplies?
- What did you deliberately leave out, and why?
- Which parts of this code would you not be comfortable shipping, and what would you fix first?

7.3 Tie-Breaking

Where two teams finish on the same total, the tie is resolved in this order:

- Higher score on System Architecture & Design.
- Higher combined score on Containerization & CI and Deployment & Production Readiness.
- Higher score on Presentation & Defence.
- Panel discretion.

8 ENGINEERING EXPECTATIONS CHECKLIST

Use this as a working checklist during the day. Every item below maps directly onto the scoring table in Section 7.1.

AREA	EXPECTATION
Design	Modular, well-organised structure with clear separation of concerns. No single-file monolith.
APIs	Consistent, predictable endpoints with sensible status codes and response shapes.
Reliability	Input validation on every entry point. Errors handled and logged rather than swallowed.
Configuration	All environment-specific values and secrets in environment variables, with an .env.example committed.
Testing	Unit tests covering core business logic. Depth over count; a handful of meaningful tests beats twenty trivial ones.
Containerization	A working Dockerfile per service and a root docker-compose.yml that brings the whole stack up in one command.
CI	A GitHub Actions workflow (or equivalent) that builds the image and runs the test suite on every push.
Deployment	The application deployed and reachable at a public URL. Poridhi VM/Cloud, AWS, Render, Railway or any platform of your choice.
Documentation	A README with an architecture overview, local setup steps, a deployment guide, and API documentation.
Version control	Frequent, meaningful commits across the whole window. Branches and pull requests if the team's workflow calls for it.

Poridhi's Dedicated Development Environment :

During the hackathon, each team will receive access to a dedicated Poridhi Lab environment. Teams can use the lab environment to launch a **Poridhi-provided AWS account** or provision a **Poridhi VM**. The available configurations can be found at this [link](#) . Applications may be deployed on either environment and exposed to a public URL using Poridhi's built-in load balancer feature.

Using the Poridhi platform for deployment is **not mandatory**. Teams may choose to deploy their application on any other deployment option mentioned above, as long as the application is publicly accessible for judging.

9 FACILITIES & SUBMISSION

9.1 What We Provide

- Venue Wi-Fi connectivity throughout the day
- Power access and workspace for every team
- Access to Poridhi's lab facilities, including the Docker and AWS lab tracks
- Access to the Puku Editor and Puku CLI for AI-assisted development
- On-site mentors available for architecture and debugging guidance
- Meals for the day: a light breakfast at check-in, a full lunch, and light evening snacks

9.2 What Participants Should Bring

- A personal laptop for each team member, with your development environment already installed and working
- Chargers, extension cords and any device-specific accessories
- Backup mobile data, in addition to the venue Wi-Fi
- A GitHub account you can access on the day
- Accounts already set up on whichever deployment platform you plan to use

SET UP BEFORE YOU ARRIVE

Installing Docker, configuring a cloud account or resolving a broken local environment on event day is time you will not get back. Arrive with a machine that can build and run a containerized project.

If you plan to deploy on AWS, complete the Poridhi AWS setup lab before the event. You already have access to it. Teams meeting AWS for the first time on hackathon day routinely lose hours to account setup, IAM and security group configuration alone.

All of this preparation is allowed and encouraged. The problem statement is revealed on the day; everything around it can be ready before you walk in.

9.3 Required Submission Deliverables

DELIVERABLE	DESCRIPTION
GitHub Repository	Complete source code developed during the hackathon. The repository must be public.
Deployment URL	Link to the live, deployed application.
README	Architecture overview, local setup instructions, deployment guide and API documentation.
Credentials	Demo login credentials, if the application requires authentication.
Submission Form	The official submission form, completed before the code freeze deadline.

Submissions are accepted up to the code freeze at 18:30. Anything committed or submitted after that timestamp is not evaluated. Submit early and update if you improve something; do not wait for the last minute.

10 PRIZES & REWARDS

The top three teams, as judged by the Poridhi.io panel, receive the following:

REWARD	DETAILS
Prize Money	Cash prizes for the top 3 teams. [AMOUNT BREAKDOWN TO BE CONFIRMED]
Puku CLI Max Plan	Worth 90,000 BDT, awarded to the top 3 teams.
Internship Opportunity	An opportunity to intern at Poridhi.io.
Certificate of Achievement	Awarded to every member of the top 3 teams.

All registered participants, regardless of placement, receive a Certificate of Participation, the Phase 1 learning content and Poridhi lab access.

RESULTS ANNOUNCEMENT

Winners are not announced on the day of the hackathon. The Poridhi.io panel reviews every submission in depth after the event.

Results will be published 1 to 2 days later on the IEEE Computer Society CUET Student Branch Chapter Facebook page. Follow the page so you do not miss the announcement.

11 SAMPLE ENGINEERING CHALLENGE

NOTE

This sample illustrates the expected depth and engineering rigour only. It is not the problem statement. The actual problem statement will be announced live at the Opening Ceremony.

Library Management System

Build a Library Management System that lets members search, borrow, return and reserve books, while giving librarians tools to manage inventory and borrowing records. The application should be modular, reliable, maintainable and ready for a production environment.

TASK	EXPECTATION
Modular Architecture	Design reusable modules for authentication, book catalog, borrowing, reservation, notification and administration.
Core Features	Implement complete member and administrator workflows, including notifications.
Reliability	Input validation, error handling, structured logging and unit tests over core logic.
Containerization	A Dockerfile per service and a working Docker Compose configuration.
CI/CD	Automated build and test workflows using GitHub Actions or an equivalent tool.
Deployment	Deploy to Poridhi VM/Cloud, AWS, Render, Railway or another platform of choice.
Monitoring & Observability (Bonus)	Implement observability using Prometheus for metrics, Grafana for dashboards, and OpenTelemetry with Jaeger/Tempo for distributed tracing. Include health/readiness endpoints and basic alerting for service failures and resource usage.
Documentation	Architecture notes, setup guide, deployment guide and API documentation.

Expected Workflow

WORKFLOW

Problem Statement → Requirement Analysis → System Design → Development → Testing → Dockerization → CI Pipeline → Deployment → Production-Ready Application → Presentation

Notice where the effort sits. Roughly the first fifth of that chain is analysis and design, the middle is implementation, and the last third is everything that makes the software shippable. Teams that skip straight from problem statement to development, and leave the tail of that chain for the final hour, consistently score lower.

12 FREQUENTLY ASKED QUESTIONS

Do I need to have attended Phase 1 to join?

No. Phase 2 is open to everyone. All registered teams receive the Phase 1 learning content and Poridhi lab access on confirmation, so you can prepare before event day.

Can I compete alone?

Yes. Solo teams are allowed and are judged on the same criteria as three-member teams. Scope your build accordingly.

I have never used Docker or deployed anything. Should I still join?

Yes, and this is the event to learn it in. Use the Phase 1 content and the Poridhi Docker and AWS labs beforehand. Partial credit is real; getting a simple service containerized and deployed scores better than a large local-only application.

What is the problem statement?

It is confidential until the Opening Ceremony. What you can prepare is your environment, your tooling, your team's workflow and your fundamentals.

Can we use starter templates and boilerplate?

Yes. Framework generators, official starter templates, UI component libraries, open-source packages and public APIs are all fine. The problem statement is only revealed on the day, so the solution is built on the day; everything around it can be prepared in advance.

Can we use AI tools like Puku, Copilot or ChatGPT?

Yes, freely. The event is Puku-friendly by design. The only condition is that you can explain your own system to the judges.

Is deployment mandatory?

It is part of the core scoring, not an optional extra. A submission that never leaves localhost loses the Deployment points and usually the Containerization points too.

Our deployment breaks near the deadline. What happens?

Submit anyway, and document the state honestly in your README. Keep a working local demo ready as a fallback for your presentation. Judges reward an honest account of a broken deploy far more than a misrepresented one.

Which cloud platform should we use?

Whichever you are fastest on. Poridhi VM/Cloud, AWS, Render and Railway are all acceptable. Free tiers are fine.

Do all team members need to present?

One member can drive the demo, but any member may be asked technical questions. Make sure nobody on the team is a stranger to the codebase.

Can we change our team after registering?

Team composition is locked at the registration deadline. Changes after that require organiser approval.

Is the registration fee refundable?

Fees are generally non-refundable once a registration is confirmed. If you have a genuine reason you can no longer attend, contact the organisers before the event and we will review it case by case.

Will food be provided?

Yes, and it is covered by your registration fee. A light breakfast at check-in, a full lunch in the afternoon, and light evening snacks later in the day.

Can we leave the venue during the day?

Short breaks are fine. The build window does not pause, and you must be present for the code freeze and your presentation slot.

Does our repository have to be public?

Yes. It must be public at the time of submission. Private repositories will not be evaluated, so make sure you flip the visibility before the code freeze.

When are the winners announced?

Not on the day. The Poridhi.io panel reviews all submissions after the event, and results are published 1 to 2 days later on the IEEECS CUET Facebook page.

What if we do not finish?

Submit what you have. An incomplete but well-engineered submission regularly places above a complete but poorly engineered one. Finishing is not the bar; engineering is.

13 CODE OF CONDUCT

Zero to Production is a learning event first. Everyone present, participants, mentors, volunteers and judges, is expected to uphold the following.

- Treat every participant, organiser, mentor and judge with respect, regardless of skill level, batch, department, gender or background.
- Harassment, discrimination, intimidation and personal abuse have no place at this event, in person or in the event channels.
- Help rather than gatekeep. Beginners are welcome here by design.
- Do not interfere with another team's work, workspace, network or equipment.
- Respect the venue, the equipment and CUET campus rules.
- Follow instructions from organisers and volunteers regarding schedule, safety and venue logistics.
- Report any concern to an organiser immediately. Reports are handled discreetly.

ENFORCEMENT

Breaches of this Code of Conduct may result in warnings, score penalties, disqualification or removal from the venue, at the organisers' discretion. Organiser decisions are final.

14 CONTACT & SUPPORT

For questions about registration, payment, eligibility or event logistics, reach out before the deadline rather than on event day.

CHANNEL	DETAILS
Registration Form	https://forms.gle/5VEtWzPyauEWQa4k9
Payment (bKash)	01837603873 — use your team name as the reference
Event & Technical Queries	Message the IEEE Computer Society CUET Student Branch Chapter page on Facebook
Results Announcement	Published on the IEEECS CUET Facebook page, 1 to 2 days after the event

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